

Electrochemical and sensing properties of AuNps-2D-MoS₂/SPCE for folic acid determination

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ABSTRACT

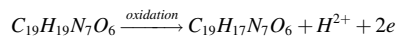
A simple and sensitive electrochemical sensor was developed for the detection of folic acid (FA) based on gold nanoparticles (AuNPs) dispersed on exfoliated MoS₂ (2D-MoS₂) nanosheets by a simple impregnation method. The prepared 2D-MoS₂ and AuNPs-MoS₂ samples were characterized by scanning electron microscopy (SEM-EDX), UV-vis, and Raman spectroscopy. Screen-printed carbon electrodes (SPCE) were modified by the 2D-MoS₂ and AuNPs-MoS₂ nanosheets and their electrochemical characteristics were investigated by means of electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) techniques and tested in the electroanalytical determination of folic acid (FA). The modified 2D-MoS₂/SPCE was found to promote the FA electrooxidation process compared to bare SPCE. Instead, with the AuNPs-MoS₂/SPCE sensor a different sensing mechanism, based on the inhibition of the Au NPs redox peaks current, has been found to be more effective. This behavior was due to the strong binding of FA on AuNPs, as probed by in situ ATR-FTIR and SERS measurements. The decrease of the anodic peak current of Au NPs with the concentration of FA provided an indirect signal which has been used here for detecting FA.

Introduction

Over recent years, thin two-dimensional materials have received great attention in the field of energy storage, catalysis, sensors, drug carriers, solar cells, and among others [1–4]. They displayed a fast-growing interest due to their peculiar properties that would play a significant role in future technology. Among these 2D materials, the transition metal disulphide (TMDS) materials show interesting features citing high-carrier mobility, large surface-to-volume ratio, and indirect-to-direct crossover when decreasing the thickness. These features are of outmost importance especially for their use in the biosensing field [5]. Indeed, compared with conventional analytical techniques, electrochemical methods display many advantages for example ease of operation and low cost.

Folic acid, known as vitamin B₉, is a water-soluble vitamin that can be found in fruits, green vegetables, animal products [6]. However, this vitamin is unstable, and it possesses an ease of degradation by pH, oxygen, UV light, and temperature. The deficiency of B₉ can affect

adversely human beings and pregnant women increasing the possibilities of diseases such as anemia, psychosis, and neural tube defects [7,8]. Various techniques can be used for the detection of FA but they are generally time consuming, high-cost, and requiring an expert for the use. Folic acid (see Scheme 1a,b showing the 3D conformer and the molecular structure) is an electroactive molecule, so its electrochemical determination, mainly based on the oxidation reaction as described in Scheme 1c, is feasible on many electrode surfaces [9,10]. Further, electrochemical methods are low-cost, simple to use, and time-saving, then their use for the determination of FA is highly promising.



In a previous paper an electrochemical sensor based on MoS₂-AuNPs dispersed on graphene nanosheets (GNS), with gold particle sizes of about 10–50 nm, and deposited onto the surface of a screen-printed electrode was reported. Due to the good synergic effect between GNS, MoS₂, and AuNPs, the composite shows excellent electrocatalytic ability for the electrooxidation of FA [11].

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Apart from the classical electrochemical scheme above illustrated, other electrochemical pathways have been explored for FA detection. For example, based on the strong ability of FA to bind many metal surfaces, adsorptive stripping techniques using a mercury electrode have been previously proposed [12,6]. In the same manner, a lead electrode was also used [13]. As both mercury and lead electrodes have the drawback of a certain degree of environmental toxicity, here we introduce more friendly electrodes.

In this work, we propose an electrochemical sensor for the detection of folic acid (FA), based on 2-D MoS₂ nanosheets. The nanosheets were synthesized through the liquid-phase exfoliation (LPE) technique which is a low-cost and easy method to produce few-layers 2D materials. Further, the 2-D MoS₂ nanosheets were coated with gold nanoparticles (AuNPs), deposited on the surface of the nanosheet by the simple impregnation method. A commercial screen-printed carbon electrode (SPCE) platform was then used to fabricate Au NPs-MoS₂/SPCE sensors for the detection of FA. We attempted the electrochemical detection of FA exploiting the well-known strong interaction between FA and gold [14]. Briefly, AuNPs nanoparticles on 2-D MoS₂ nanosheets can serve as attachment sites for FA. The steric/electronic hindrance induced by the formation of the FA-Au complex prevents electrons from reaching the surface of the electrode and consequently decreases the current, which is thus exploited for the development of an electroanalytical sensor. A similar procedure, where AuNPs have been used as electrochemical and plasmonic substrates for the detection of FA, has been reported in previous papers [15–18]. Rather than using the expensive procedure and precursors previously reported in the cited papers, we used low-cost, accurate, real-time method, but, to our knowledge, this represents the first attempt for detecting FA by the described sensing mechanism with disposable screen-printed electrodes.

Experimental

Synthesis of 2D-MoS₂ nanosheets

2D-MoS₂ nanosheets were prepared by surfactant-assisted liquid-phase exfoliation of MoS₂ powder in a watery solution of sodium cholate (SC – 1.5 mg/mL), whose stabilizing action prevents the MoS₂ nanosheets aggregation [19–21]. MoS₂ powder was dispersed in the SC solution at the final concentration of 5 mg/mL. Afterwards the dispersion was sonicated with Branson 250 sonifier mounting a flat tip (3/16" diameter), and operating at 50 % of amplitude for 30 min with an ultrasonics time-on/time-off ratio of 1:1 for every second. Samples were kept in an ice bath to reduce detrimental heating effects during sonication. After the sonication process, the dispersions were left to decant in a flask overnight. After the rest period the supernatant was separated from the sediment and centrifuged at 1500 rpm for 90 min (ALC 4235, swinging rotor). The top part of the centrifuged sample (rich in few-layer nanosheets) was extracted and used in our experiments.

The hybrid sample combining AuNPs and MoS₂ nanosheets (2D-MoS₂) was fabricated through the impregnation method [22]. In

particular we mixed, at room temperature, 0.2 ml of a HAuCl₄ solution (5 mM) with 0.2 ml of a 2D-MoS₂ suspension, keeping the final sample under stirring for 1 min. In these conditions, spontaneous electron transfer from MoS₂ to gold ions occurs leading to the formation of Au nanoparticles [22].

Characterization

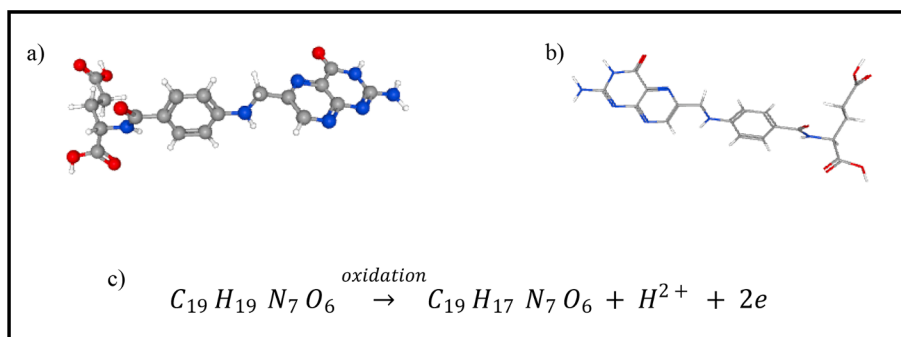
SEM images were acquired with a Zeiss CrossBeam 540, while UV–vis spectra were recorded in the 250–900 nm wavelength range with a Perkin Elmer Lambda 25 UV–vis setup using 1 cm length sample cell. Micro-Raman spectroscopy experiments were carried out on MoS₂ flakes at room temperature with a Horiba Labram HR800 confocal spectrometer (Horiba JobinYvon) coupled with an Argon ion laser emitting at 514.5 nm (Spectra Physics). The laser beam is focused on the sample surface using a 100X objective (NA 0.9, WD 0.2 mm) and the scattered light is collected through the same objective in a backscattering configuration. The Raman signal is then dispersed by a 1800 l/mm grating and detected by a Peltier-cooled silicon CCD (Synapse, Horiba Jobin-Yvon). Spectra of the MoS₂ drop on glass were acquired using a laser power of 9mW, integration time of 20 s, with up to 10 accumulations. SERS characterizations were performed using an Horiba Xplora coupled with a 785 nm diode laser with a power at the sample of 2 mW, integration times of 20 s and 3 accumulation. The Raman signal is then dispersed by a 1200 l/mm grating and detected by a Peltier-cooled silicon CCD (Syncerity, Horiba Jobin-Yvon). Fourier transform infrared spectroscopy (FT-IR) spectra were collected by a Perkin Elmer Spectrum 100 spectrometer equipped with a universal ATR sampling accessory.

Fabrication of Au-MoS₂/SPCE sensor

Screen-printed carbon electrodes (SPCEs) were purchased from DropSens, Spain. The platform used was: DRP-100 (named SPCE), constituted of a 4-mm diameter carbon working electrode, a silver pseudo reference electrode, and a carbon auxiliary electrode. To modify the bare SPCE with 2D-MoS₂ or Au NPs-MoS₂ nanosheets, 50 µl of an aqueous dispersion containing these nanosheets, were deposited on the carbon working electrode and allowed to dry at room temperature until further use. These sensors are named MoS₂/SPCE and Au NPs-MoS₂/SPCE, respectively.

Electrochemical measurements

Electrochemical analyses were performed by using a DropSens µStat 400 Potentiostat empowered by Dropview 8400 software for data acquisition. Electrochemical impedance spectroscopy (EIS) tests were performed by using an Autolab PGSTAT204 system in ferrocyanide solution ([Fe(CN)₆]^{4-/3-}) at frequency range 10⁵–10² Hz. The sensors were characterized by cyclic voltammetry (CV) tests in aerated 1 M PBS as electrolyte. CV tests were carried out at a scan rate of 50 mV/s. All experiments were performed at room temperature.



Scheme 1. a) 3D conformer and b) molecular structure of folic acid; c) oxidation reaction of FA.

Results and discussions

Morphological, microstructural, and electrical measurements

The prepared 2D-MoS₂ sample was first investigated for evaluating its morphological and microstructural characteristics. UV-vis measurements were performed on solutions diluted at 1:10 v/v to minimize the influence of multiple scattering. Under these conditions information on the samples' concentrations can be obtained by applying Beer's law. Fig. 1 reports the UV-vis spectrum, showing the A- and B- excitonic peaks, respectively at ~676 nm and ~613 nm, together with the C-excitonic band at 467 nm, superposed to a scattering background, and consistent with what was expected for the 2H polytype of MoS₂. Moreover, the direct transition of excitons from VB to CB results in peaks located at 382 nm, named D excitonic band.

The average nanosheets size $\langle L \rangle$, concentration, and average number of layers $\langle N \rangle$, are parameters characterizing the 2D- exfoliated nanomaterials that can be extracted from empirical formulas [23]. From the ratio between the local minimum extinction intensity at 350 nm (Ext350) and the extinction value at the B-exciton peak wavelength (ExtB), we can calculate $\langle L \rangle$:

$$L (\mu\text{m}) = \frac{3.5 * \text{Ext}_B / \text{Ext}_{350} - 0.14}{11.5 - \text{Ext}_B / \text{Ext}_{350}} \quad (1)$$

Based on the extinction on the empirical absorption coefficient $\epsilon_{350} = 69 \text{ L g}^{-1} \text{ cm}^{-1}$, the concentration of these dispersions is computed as [23]:

$$C = \frac{1}{l} \text{Ext}_{350} / \epsilon_{350} = -\frac{1}{\epsilon_{350} l} \text{Log}_{10} T(\lambda = 350 \text{ nm}) \quad (2)$$

where l is the length of cuvette containing the solution (1 cm in our case) and $T(\lambda = 350 \text{ nm})$ the transmittance at the local minimum at 350 nm. From the position of the A-exciton λ_A (in nm), $\langle N \rangle$ can be estimated as follows:

$$\langle N \rangle = 2.3 * 10^{36} e^{-54888/\lambda_A} \quad (3)$$

From the data in Fig. 1, the average number of layers $\langle N \rangle \sim 12$, with an average length $\langle L \rangle \sim 200 \text{ nm}$ and concentration in the order of 32 $\mu\text{g/mL}$.

Raman spectroscopy

Fig. 2 shows the Raman fingerprint of MoS₂ nanosheets which is dominated by two sharp peaks, the E_{2g}^1 (385 cm^{-1}) and the A_{1g} (410 cm^{-1}). The first mode is related to the in-plane vibration of S with M atom in-plane while the second mode represents the out of plane S

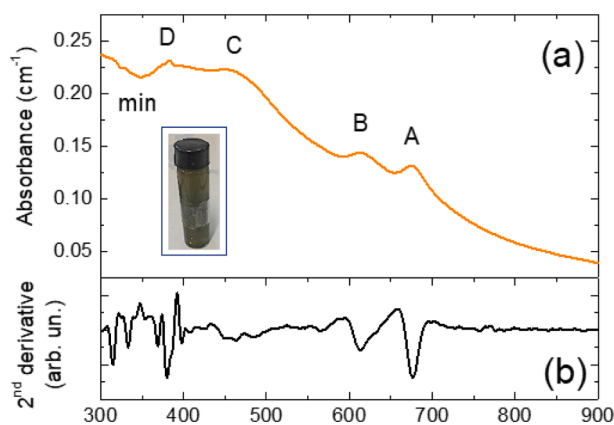


Fig. 1. (a) Extinction spectrum of the 2D-MoS₂ dispersion (diluted 1:10 v/v). (b) Second derivative of the extinction signal. Inset: Picture of the MoS₂ nanosheets suspension (non-diluted solution).

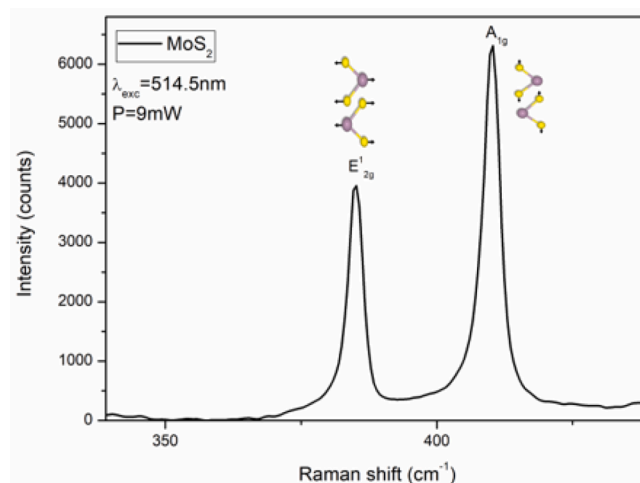


Fig. 2. Raman spectrum of MoS₂ at room temperature under 514.5 nm laser line. Inset: atomic displacement of active (E_{2g}^1 , A_{1g}).

vibration (inset Fig. 2) [24]. The number of layers of MoS₂ is sensitive to the in-plane and out of plane modes wavelength, then, the difference between these modes, $\Delta\omega (E_{2g}^1 - A_{1g})$, can be an indicator for the number of layers contained in the nanosheets [25]. For instance, passing from bulk to single layer the $\Delta\omega$ changes from 26 cm^{-1} to 17.5 cm^{-1} [26]. The difference between the two modes in our case is equal to 25 cm^{-1} , corresponding to more than 6 layers, which is in agreement with previous investigations [27]. Using the relationship $\Delta\omega (A_{1g} - E_{2g}^1) = 25.8 - 8.4/N$, we can also determine number of layers, that in our case is equal to 10 [28,29]. The average number of layers estimated by Raman spectroscopy ($N \sim 10$) is slightly lower than that retrieved by UV-vis ($N \sim 12$). However, some small discrepancy can be acceptable since these two calculations are based on different techniques.

SEM characterization

SEM image reported in Fig. 3a,b shows the morphology of the bare SPCE and the as prepared 2D-MoS₂ nanosheets dispersed on the SPCE surface, respectively.

SEM images from the Au modified-MoS₂ deposited layer on the SPCE electrochemical platform, are shown in Fig. 4. At low magnification (Fig. 4a), AuNPs are seen dispersed on the surface of the MoS₂/SPCE, forming aggregates. Gold nanoparticles with size ranging from 80 to 200 nm, can be clearly seen at higher magnification (Fig. 4b). This confirms previous data in ref. [22], where is reported that a redox reaction occurs between MoS₂ and gold ions, allowing spontaneous electron transfer from MoS₂ to gold ions and leading to the formation of Au nanoparticles on the surface of chemically exfoliated TMDC sheets. The density and size of the gold nanoparticles decorating the MoS₂ sheets can be easily tuned by controlling the concentration of HAuCl₄. Further, the size distribution has been demonstrated to be maintained relatively narrow, without the use of any stabilizer.

The presence of the expected elements is confirmed from the EDX spectrum in Fig. 4c: Au (wt%=71), C (wt%=24), Mo (wt%=4), and S (wt %=0.3). The presence of chlorine (Cl) element may be due to the contamination of this ion by HAuCl₄ used as precursor material for Au during the preparation of Au-MoS₂ nanosheets.

Electrochemical characterization of AuNPs-MoS₂/SPCE

The redox probe $[\text{Fe}(\text{CN})_6]^{3-/4-}$ has been used for the study of the electrochemical characteristics of SPCE, MoS₂/SPCE and AuNPs MoS₂/SPCE, respectively, under the following conditions: scan rate 0.05 V/s and the potential window between -0.2 V and -0.6 V .

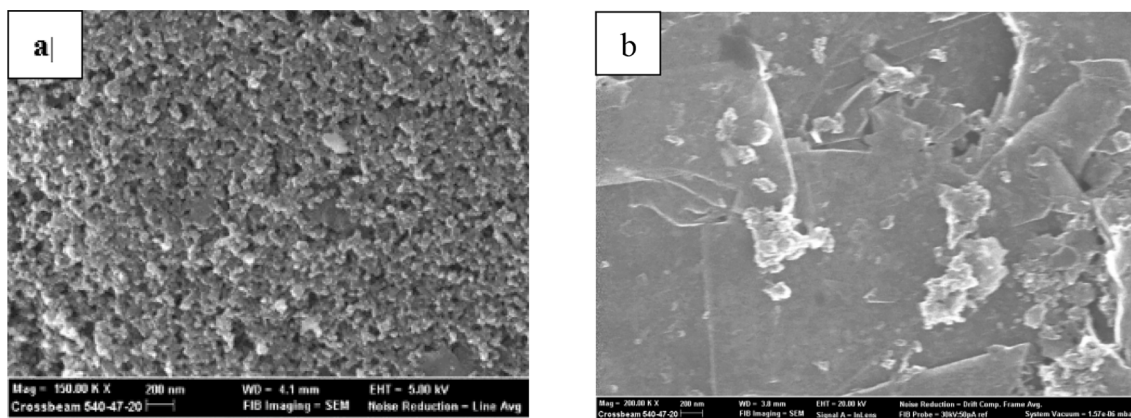


Fig. 3. SEM images showing the morphology of: a) bare SPCE; b) 2D-MoS₂ nanosheets over SPCE surface.

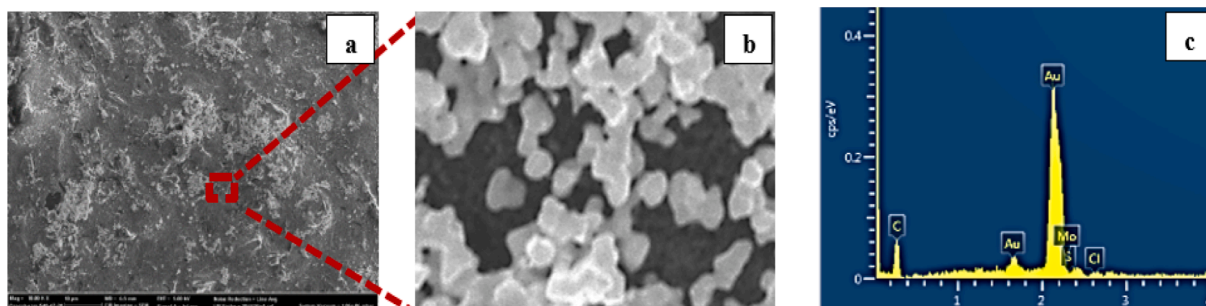


Fig. 4. a, b) SEM image at different magnification of the AuNPs-MoS₂/SPCE surface; c) EDX spectrum.

As noted in Fig. 5, MoS₂/SPCE shows only a minimal improvement in the current response with respect to bare SPCE. Instead, using Au NPs-MoS₂/SPCE revealed an enhancement in anodic and cathodic currents (I_{pa} and I_{pc}) compared with bare and modified MoS₂/SPCE. This finding appears to be due to the higher surface area of the modified electrode.

Based on the equation of Randles-Sevcik, the active surface area (SA) of unmodified and modified electrodes were calculated [30]:

$$I_{pa} = 2.69 \times 10^5 SAD^{1/2} n^{3/2} \nu^{1/2} C$$

where I_{pa} is the anodic peak current in μA , D is the diffusion coefficient

in $cm^2 s^{-1}$ ($7.6 \times 10^{-6} cm^2 s^{-1}$), C is the concentration of the analyte in $mol cm^{-3}$. The number of electrons involved in the oxidation reaction is n and the scan rate is ν . Calculation indicated that the active surface area of AuNPs-MoS₂/SPCE is increased by 12 % compared to that of the unmodified SPCE, confirming the expected results. We have further characterized the electrochemical response of the various electrodes employing electrochemical impedance spectroscopy (EIS). The Nyquist plot collected in $[Fe(CN)_6]^{3-/4-}$ solution is reported in Fig. 6a. In the case of bare SPCE, we observed a semicircle in high-frequencies and a linear segment in low-frequencies. Through the analysis of the equivalent circuit of the electrochemical cell (Fig. 6.b), we can calculate R_s , R_{CT} , C_{dl} , and Z_w standing for electrolyte resistance, charge transfer resistance, double-layer capacitance, and Warburg impedance, respectively (see Table 1). For the bare SPCE, R_{CT} was equal to $\sim 3707 \Omega$. The significant change in R_{CT} value is attributed to the modification of the charge-transfer rate across the modified interface. This value becomes $19.6 k\Omega$ and $19.3 k\Omega$ for MoS₂/SPCE and Au-MoS₂/SPCE, respectively.

In summary, the electrochemical characterization above reported indicate that improvement of the electrochemical properties of the Au NPs-MoS₂ composite, is not only due to a different surface area but the presence of Au NPs as the electroactive sites should be considered as prominent.

Electroanalytical determination of FA on AuNPs-MoS₂/SPCE

Preliminary tests in 0.01 M phosphate buffer solution (PBS, pH = 7.4) with bare SPCE, modified MoS₂/SPCE, and both Au NPs MoS₂/SPCE were first performed in the potential range 0.0 V to 1.0 V and at 0.05 V/s scan rate (Fig. 7). Test on bare SPCE and MoS₂/SPCE in PBS showed a smooth cycle without any faradaic peak in the potential window investigated. However, on the MoS₂/SPCE the CV cycle is much larger, due to the occurrence of higher capacitive effects. This finding is intuitively linked to the expected higher surface area of this modified SPCE

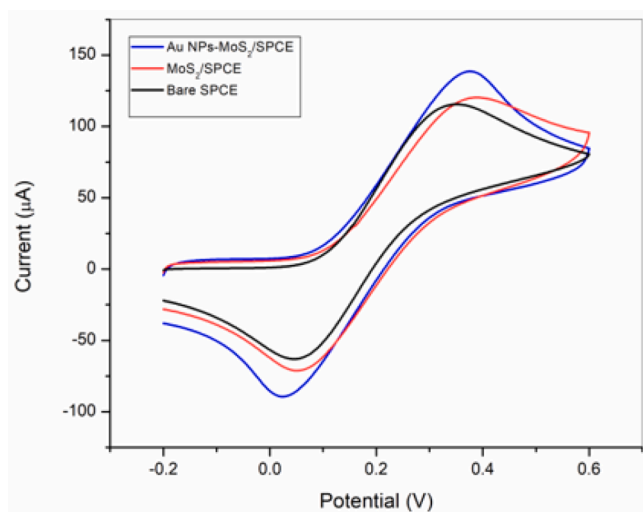


Fig. 5. CV in $[Fe(CN)_6]^{3-/4-}$ solution for the SPCE, MoS₂/SPCE and AuNPs MoS₂/SPCE.

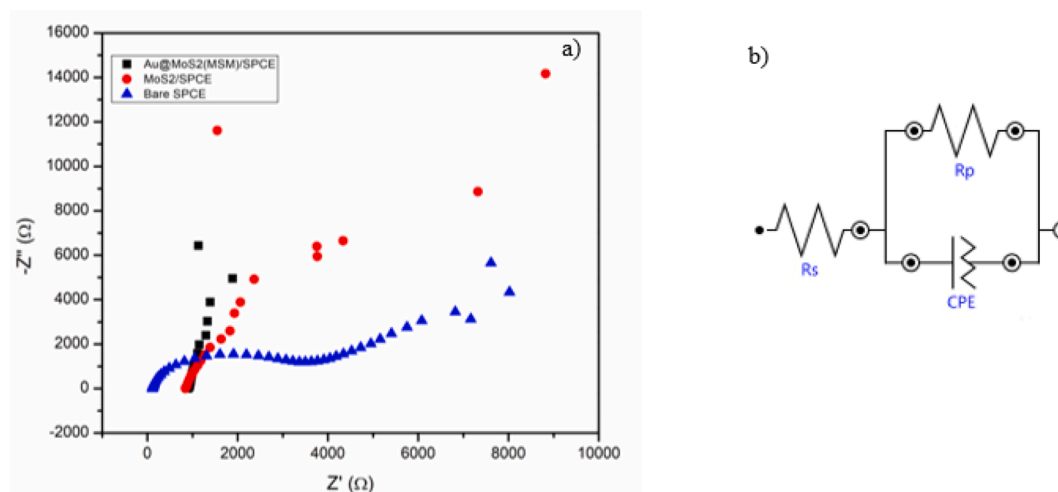


Fig. 6. (a) Nyquist plot of bare SPCE, MoS₂/SPCE and Au NPs-MoS₂/SPCE. (b) equivalent circuit.

Table 1

Randles equivalent circuit parameters; R_s and R_{CT} .

Electrode nature	R_s (Ω)	R_{CT} (K Ω)
Bare SPCE	92.855	3707.1
MoS ₂ /SPCE	909.08	19,693
Au@MoS ₂ (MSM)/SPCE	1015.5	19,333

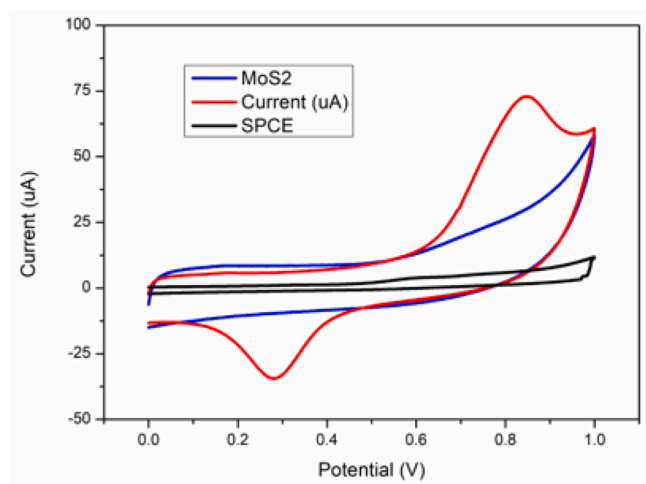


Fig. 7. CV of bare SPCE, MoS₂/SPCE, and Au NPs MoS₂/SPCE in PBS solution at 50 mV s⁻¹.

with respect to the bare one. Vice versa, the Au NPs-MoS₂/SPCE sensor exhibits a pair of peaks, at +0.27 V and +0.85 V, attributed to redox cycle Au_{ox}/Au_{red} which dominates the CV cycle, confirming the successful AuNPs formation.

With the objective to develop a method for the electroanalytical determination of FA based on the described sensors, they were then investigated by Linear Sweep Voltammetry (LSV) in the presence of different folic acid concentrations (Fig. 8a). On the bare SPCE a weak anodic peak was observed to appear in the presence of FA at ~0.60 V. Compared with bare SPCE, the MoS₂/SPCE (reported in the inset of Fig. 8a) showed an anodic peak at ~0.52 V. The shift to lower potential, along an increase of the anodic peak current (Fig. 8b), was an indication of the good electrocatalytic ability of MoS₂ nanosheets for folic acid electrooxidation.

Tests with different FA concentrations were then performed on the

AuNPs-MoS₂/SPCE sensor. We noted that by using this sensor the anodic peak at around 0.5 V is almost negligible. On the other hand, we observed a strong decrease of the intensity of the gold redox peaks with increasing the concentration of FA (Fig. 9a). Further, after each addition of FA results in a systematic decrease of redox current of both anodic and cathodic gold peaks. What occurred at the anodic peak on the addition of folic acid is much clearer, subtracting the baseline in absence of FA and reporting the variation associated (see Fig. 9b). The change of anodic peak current observed on AuNPs-MoS₂/SPCE with FA concentration is shown in Fig. 9c and compared with the variation of the current of the electrooxidation peak of FA observed on the bare and MoS₂/SPCE sensors. The AuNPs-MoS₂/SPCE sensor clearly exhibits a larger variation than the signals showed by the other investigated sensors. Based on the above reported graphs, two linear regimes were noted. The sensitivity, computed in the concentration range 0 to 20 μ M is equal to 20.636 μ A μ M⁻¹ cm⁻² with a limit of detection (LoD) calculated using the formula: $3\sigma/S$, where σ is the slope and S is the standard deviation, of 1.04 μ M, while in the higher concentration range between 20 and 100 μ M the sensitivity is comparatively lower and is equal to 3.949 μ A μ M⁻¹ cm⁻².

Sensing mechanism

The different mechanisms we observed on the SPCE and MoS₂/SPCE from one hand, and the Au NPs-MoS₂/SPCE on the other hand, can be explained based on the different interaction of the folic acid analyte with the sensing layer electrode and specifically if Au is present or not.

In the case of the SPCE and MoS₂/SPCE, i.e. when Au is not present, the carbon surface of the first sensor and the MoS₂ terminated surface of the modified sensor, behave as the activated surfaces for the electro-oxidation of FA, following the reaction pathway shown in Scheme 1a. MoS₂/SPCE has a larger surface area than the bare SPCE and this makes it more sensitive for FA sensing.

The results reported with the Au NPs-MoS₂/SPCE, suggests that this mechanism is not operative. We explain the observed finding on the assumption that the surface of the Au NPs-MoS₂/SPCE is covered by the folic acid molecules strongly adsorbed on the Au NPs (see Scheme 2b). The adsorbed FA molecules act as a poisoning layer which does not permit the electrooxidation of FA as noted for the SPCE and MoS₂/SPCE and, even at very low concentration where the full surface coverage could not be reached, this limits the oxidation/reduction redox pathway of Au NPs.

The high binding affinity of FA for the Au surface was well described in previous papers. Castillo et al. [31] demonstrated by SERS, Raman analysis, and DFT calculations, that FA interacts with gold nanopillars mainly through the nitrogen in the pterin moiety (see red circle in

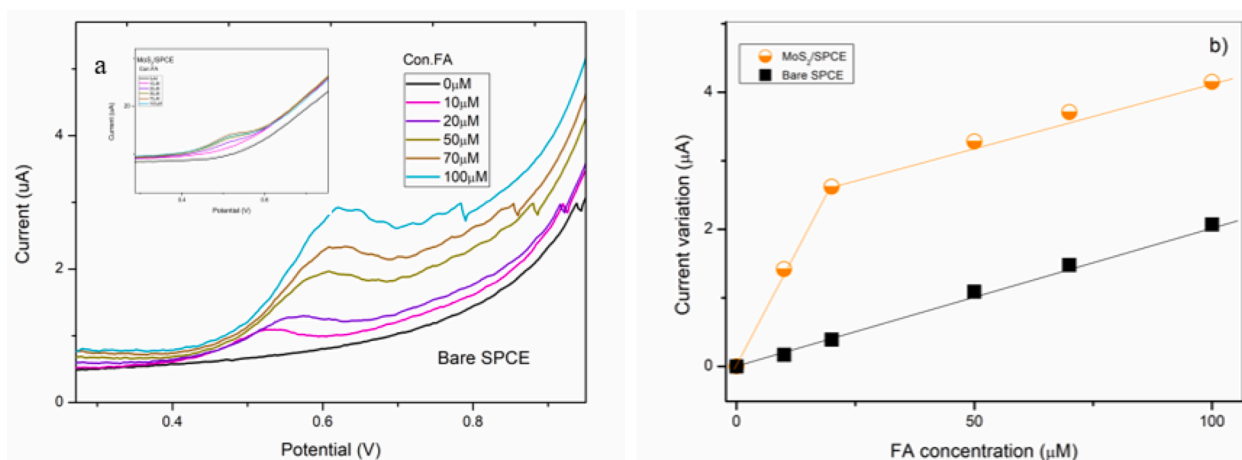


Fig. 8. a) LSV measurement and (b) calibration curve of FA on the bare SPCE and MoS₂/SPCE.

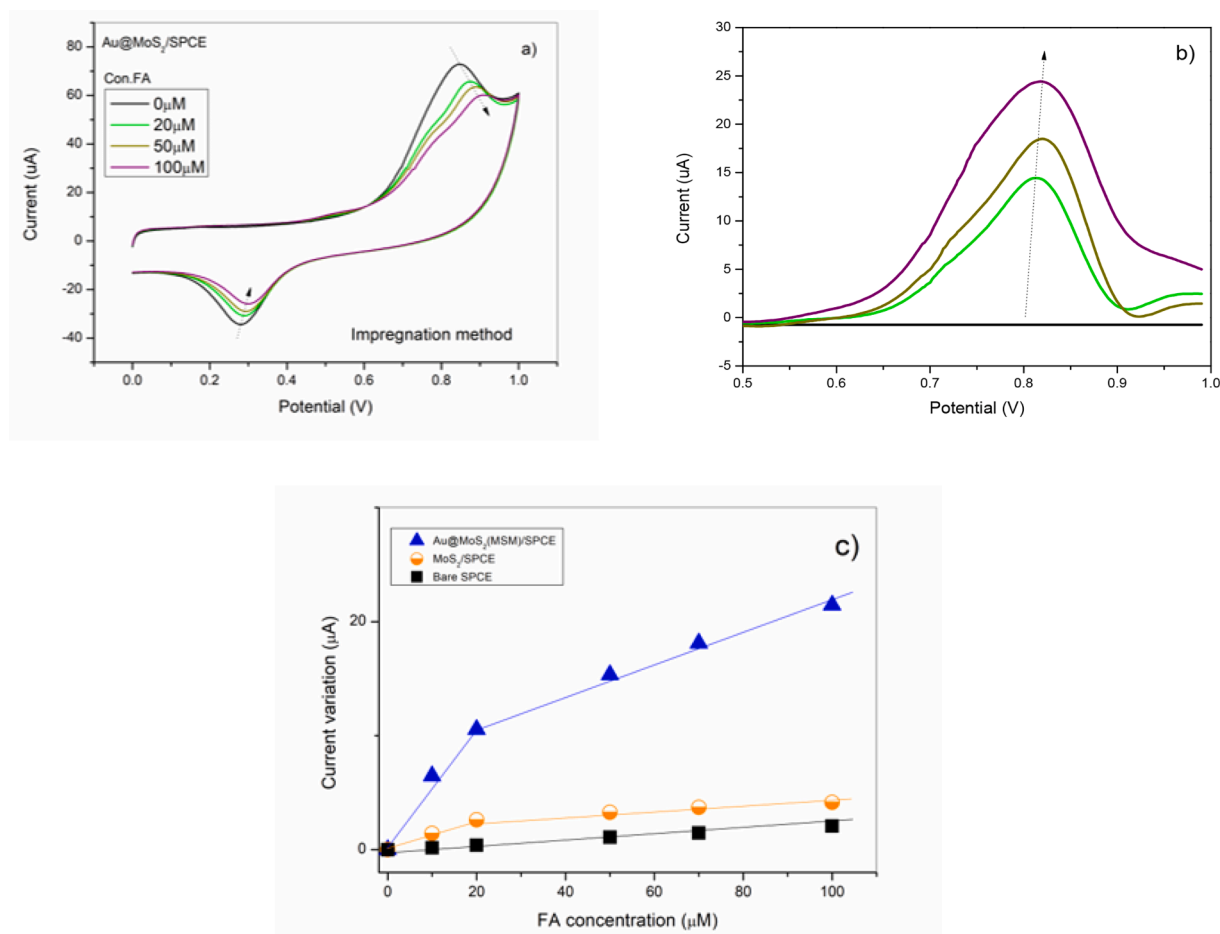
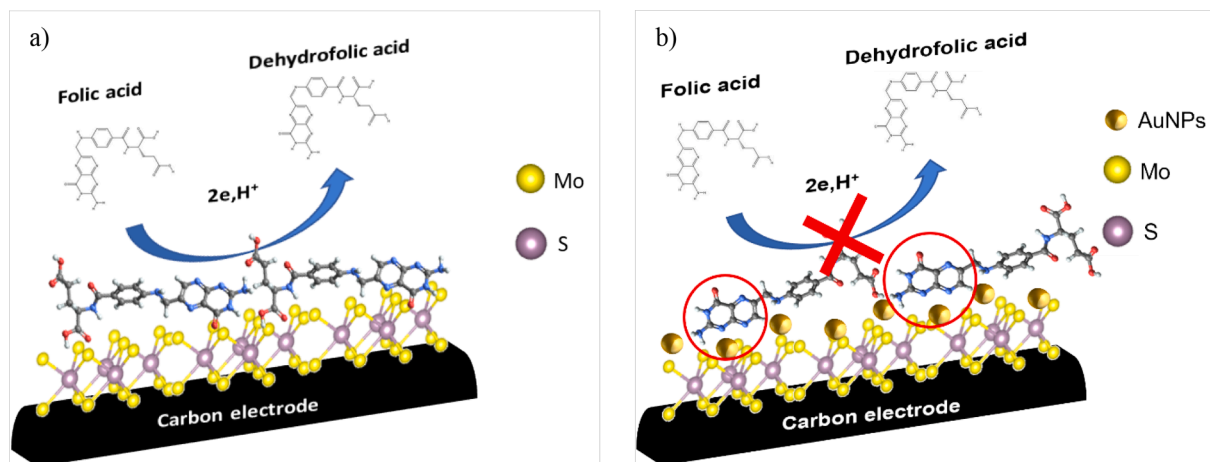


Fig. 9. (a) CV measurement at different concentration of FA on AuNPs-MoS₂/SPCE; (b) LSV measurement at different concentration of FA on AuNPs-MoS₂/SPCE with subtraction of the baseline (c) Calibration curve of AuNPs-MoS₂/SPCE. Calibration curve of FA on SPCE and MoS₂/SPCE are also shown for comparison.

Scheme 2b). However, it cannot be excluded that FA molecules are joined mainly by the carboxylic groups to the metal surface of the Au, leaving out the pterin ring. As probed by FT-IR, Majidi et al., proposed that FA can interact with the surface of the AuNP through the carboxylate groups of glutamic acid portion and the amine group of pterin moiety [32].

Anyway, when FA binds to Au NPs, the number of gold surface sites which remains free decrease. Thus, during LSV, the FA adsorbed layer

prevents the stripping of Au, resulting in the observed reduction of current. A similar mechanism has been reported for the detection of glutathione (GSH) on Au NPS on TiO₂ nanotubes [33]. On adding GSH, the redox current is decreased due to the binding of GSH to Au NPs. However, compared to our sensors, the redox current due to Au NPs on TiO₂ nanotubes decreased only after incremental addition of GSH starting from 1 mM GSH, whereas we can detect FA in the micromolar or less range.



Scheme 2. Schematic representation of adsorption/reaction processes occurring in the presence of FA on the surface of: a) MoS₂/SPCE and b) Au NPs-MoS₂/SPCE. The red circle marks the pterin portion of FA.

To highlight the strong interaction between the FA with the surface of AuNPs, the AuNPs-MoS₂/SPCE was investigated through ATR-FTIR and SERS spectroscopies. It is well known that in the presence of an interaction between the probe molecules and gold nanoparticles, a strong enhancement of the signal is detected [34]. In Fig. 10 are shown the ATR-FTIR spectra of free FA and ones collected on bare SPCE, MoS₂/SPCE, and AuNPs-MoS₂/SPCE. The ATR-FTIR spectrum of pure folic acid revealed the characteristic absorption peak at 1692 cm⁻¹, which corresponds to CO stretching from -COOH, and 1607 cm⁻¹, 1475 cm⁻¹, which correspond to the phenyl and pterin rings.

On bare SPCE and MoS₂/SPCE, the folic acid signature is barely discernible from the background. Instead, with AuNPs-MoS₂/SPCE, the folic acid spectrum is well evidenced. This suggests that folic acid interaction occurs with AuNPs presents on the surface of AuNPs-MoS₂/SPCE. Additionally, the absence of the peak due to skeletal vibration of the pterin ring at 1508 cm⁻¹ in the spectrum of this latter sample could indicate that AuNPs perhaps interact strongly with the pterin portion of FA.

Raman analysis performed on FA adsorbed on the bare SPCE didn't reveal the characteristic vibrational modes of the target molecule (Fig. 11 – black line), while for MoS₂/SPCE, some characteristic peaks of FA were observed in the low-frequency region together with the ones of MoS₂ (Fig. 11 – blue line). On the other side SERS analysis on AuNPs MoS₂/SPCE, highlights strong vibrational peaks that can be related to

the FA, which indeed is characterized by a strong chemisorption on Au NPs, in agreement with the results of Kokaislová and Matějka [35].

In particular, the SERS signal on this sample presents two vibrational modes belonging to MoS₂ NS at 237 cm⁻¹ (LA(M)) and 468 cm⁻¹ (2LA) and others peaks in the fingerprint region that can assigned to FA [29]. Indeed, three prominent vibrational modes of FA were observed at 690 cm⁻¹, 1238 cm⁻¹, and 1574 cm⁻¹. The first one assigned to either $\nu_s(\text{C}=\text{N})$ pteridine(pt) group or $\nu_{as}(\text{C}=\text{C})$ p-aminobenzoic acid (paba) [33]. The second and the third vibrational modes are assigned to two different couplings: $\rho(\text{NH})(\text{OH}) + \omega(\text{CH}_2)$ of glutamic acid (Glu) and $\nu(\text{CN}-\text{NH}_2) + \nu(\text{CC}-\text{CN})_{\text{pt}}$ [35]. The vibrational mode at 1574 cm⁻¹ can assigned either to $\nu_s(\text{NH}_2) + \nu_{as}(\text{c}=\text{n})_{\text{p t}}$ or to $\nu_s(\text{c}-\text{c}) + \rho(\text{CH})(\text{Paba})$ [35].

In the low-frequency region, the peak at 615 cm⁻¹ can be due either to the NH₂ out of plane stretching or to the in-plane OH stretching of the pteridine group [18]. As far as we know, the vibrational mode at 761 cm⁻¹ is not assigned yet while the vibrational mode observed at 817 cm⁻¹ can be assigned either to the stretching of C–N deformation of the paba group or to the CNO bending and the (OH)(CH₂) stretching modes of glutamic acid coupling [18]. The rocking mode of CH₂ in the p-aminobenzoic acid is observed at 959 cm⁻¹ while OH rocking deformation of the pteridine group is at 996 cm⁻¹ [18,35]. A coupling between rocking of (NH)(OH) mode and CH₂ wagging mode of glutamic acid is observed at 1024 cm⁻¹ [35]. The peak at 1052 cm⁻¹ is due to three different modes: C–N deformation stretching of p-aminobenzoic acid group, (NH₂)(CH)(OH) deformation bending of pteridine group, and C–C stretching deformation of glutamic acid [18].

In the high-frequency region, the peak at 1195 cm⁻¹ is due to the bending mode of OH in the pteridine group coupled with CH₂ in the glutamic acid group [36]. All the observed vibrational modes are listed in Table 2. Castillo et al suggest that the absence of vibrational modes located at 661 cm⁻¹ and 1489 cm⁻¹ reveals significant information. Indeed, the absence of the first peak indicates the strong interaction between the benzene ring of paba and AuNPs while the second suggests the indirect contact between AuNPs and glutamic acid [36]. The huge differences in SERS spectra of FA compared to FA signal detected on MoS₂/SPCE is due to the strong interaction occurring between FA molecules and the gold surface, which foster the creation of a surface complex [37]. In fact, the pteridine can be considered as an anchor site since vibrational mode located at 1574 cm⁻¹ is shifted and enhanced, suggesting this group is very close to the gold surface.

Conclusion

An easy electrochemical method for the detection of the FA was

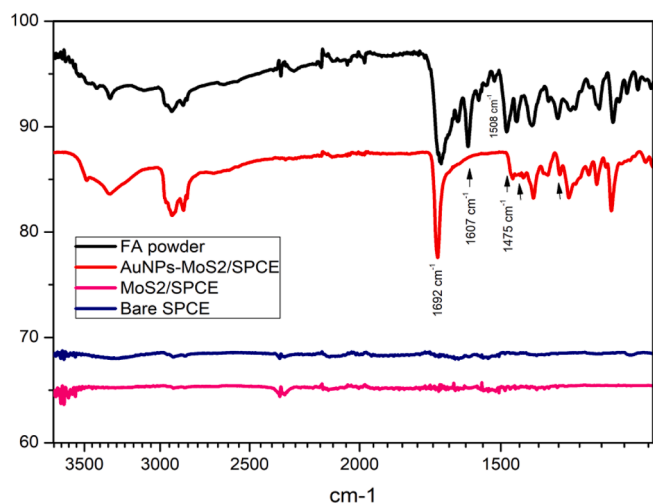


Fig. 10. FT-IR spectra acquired from FA deposited from a solution 10⁻⁴ M on the investigated electrodes.

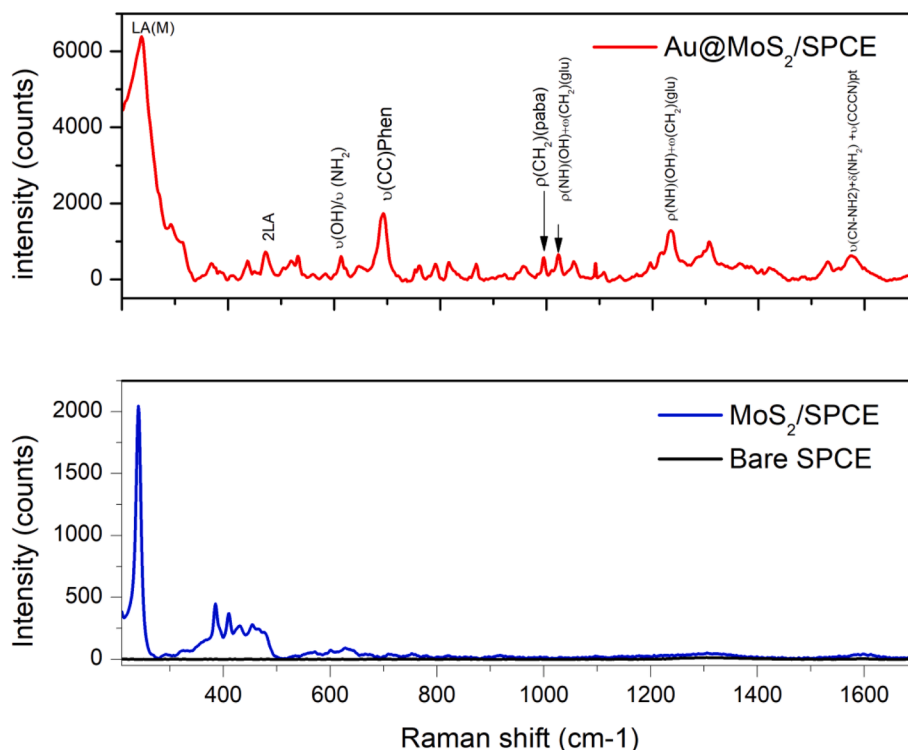


Fig. 11. SERS spectra acquired from FA deposited from a solution 10^{-4} M on the investigated electrodes.

Table 2
Assignment of SERS vibrational mode.

Wavelength (cm^{-1})	Assignment	Ref.
237	LA(M); MoS ₂	[29]
468	2LA; MoS ₂	[29]
615	$\rho(\text{OH})(\text{pt})$ IP DEF; $\rho(\text{NH}_2)(\text{pt})$ OOP	[18]
761	**	[18]
817	$\delta(\text{CNO})\rho(\text{OH})(\text{CH}_2)(\text{Glu}); \nu(\text{C}-\text{N})(\text{paba})\text{def}$	[18]
959	$\rho(\text{CH}_2)(\text{paba})$	[35]
996	$\nu(\text{C}-\text{C}) \rho(\text{CH})(\text{paba})\text{def}; \rho(\text{OH})(\text{pt})\text{def};$	[18]
1024	$\omega(\text{CH}_2)(\text{Glu}) + \rho(\text{NH})(\text{OH})$	[35]
1052	$\nu(\text{C}-\text{C})(\text{paba}); \rho(\text{NH}_2)(\text{CH})(\text{OH})(\text{pt})$	[18]
1195	$\text{def}; \nu(\text{C}-\text{C})(\text{glu})\text{def}$	[36]
1238	$(\text{CH}_2)(\text{Glu}) + \delta(\text{OH})(\text{pt})$	[35]
1574	$\rho(\text{NH})(\text{OH}) + \omega(\text{CH}_2)(\text{Glu}); \nu(\text{CN}-\text{NH}_2) + \nu(\text{CC}-\text{CN})\text{pt}$	[35]
1574	$s(\text{NH}_2) + \nu_{\text{as}}(\text{C}=\text{N})(\text{pt})$ or $\nu_s(\text{C}-\text{C}) + \rho(\text{CH})(\text{paba})$	[35]

developed based on the inhibition of the electrochemical activity of Au NPs deposited on exfoliated 2D-MoS₂ nanosheets when the gold surface is interacting with FA molecules. Indeed, differently from the 2D-MoS₂/SPCE, which was found to promote the FA electro-oxidation process, on the AuNPs-MoS₂/SPCE the surface is self-inactivated by FA due to binding between FA and AuNPs and consequently the FA electro-oxidation process is hindered. The FA-AuNPs binding was probed on the different electrodes by in situ ATR-FTIR and SERS measurements. This binding lead also to a remarkable decrease of the redox activity of AuNPs, which is reflected in the diminution of the intensity of anodic/cathodic Au couple, providing thus an indirect signal which has been used here for detecting FA.

CRediT authorship contribution statement

Khoulood Abid: Conceptualization, Investigation, Methodology, Data curation, Writing – original draft. **Rayhane Zribi:** Investigation, Methodology. **Ramzi Maalej:** Supervision. **Antonino Foti:** Supervision, Data curation. **Amani Khaskhoussi:** Investigation. **Pietro Giuseppe**

Gucciardi: Supervision, Data curation. **Giovanni Neri:** Supervision, Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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